

MicroCode: live, portable programming for children via robotics

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ABSTRACT

The BBC micro:bit is a popular tool in education for teaching coding, but typically requires a host computer and internet access, limiting its accessibility. MicroCode addresses this by enabling portable programming directly on the micro:bit using a battery-powered accessory with an OLED screen and navigation buttons. This system utilises a simple, handheld graphical tile-based programming paradigm, yet supports complex programs with features including conditional execution and variables, providing immediate feedback through live programming. This paper illustrates how early studies have received a positive reception from children and educators, especially when paired with robotics as an application domain. Plans for future work aim to extend the reach of Microcode by providing more tangible digital learning opportunities to pre-literate children and communities around the world where access to mains power and internet are scarce.

CCS CONCEPTS

• **Computer systems organization** → *Sensors and actuators*; • **Human-centered computing** → **User interface programming**; **Graphical user interfaces**.

KEYWORDS

micro:bit, MicroCode, programming, live, portable, robotics

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1 INTRODUCTION AND RELATED WORK

Physical computing devices are re-programmable computing systems that can interact with their physical environment [10, 13] and are popular in education as a tool to teach children about computing including how to code. According to Vee [19] learning to code is comparable to and just as essential as learning to read and write.

The concept of “Coding as literacy” [2] recognizes the importance of digital literacy as a lifelong learning skill and an inclusion factor for a wealthy digital society. [14]

The BBC micro:bit, Figure 1a, is one such device that has grown in popularity in recent years [1] – it is small with embedded sensors, can be battery-powered and coded in multiple programming languages including block and text-based languages. It has been used in schools to measure energy use [18], as a data collection tool [17] and to teach children about machine learning [3].

One disadvantage of many physical computing devices, including the micro:bit, is that they need a separate host computer – such as a desktop, laptop, tablet or smartphone – to program them. In many cases, this host computer must also be connected to the Internet to load a web-based programming environment such as MakeCode [1, 6]. The need for a host computer makes many physical computing devices and robotics inaccessible to various communities around the world, notably those without reliable Internet, mains electricity or resources such as computers, tablets or mobile phones.

We use the term *portable programming* when a physical computing device does not require a connection to a host computer for coding. Examples of portable programming exist in the literature including the Bee-Bot mobile robot [8], a standalone robot bee with buttons on its back. Children press the buttons to define a sequence of movements for the bee; empowering very young, pre-literate children. However with no screen, there is no reminder of what buttons were pressed. This increases cognitive load on the children. Cubetto [5] is a robot where children use physical arrow-shaped tiles and place them in order in a box. When the children press “Go”, a separate robot on a mat moves accordingly, a square at a time.

Both BeeBot and Cubetto also provide immediate feedback; as soon as a button is pressed, something happens. This is known as *live programming* and is known to be beneficial for students, and particularly novices [11], as they feel in control and can directly see the effect of their actions. This can, in turn, improve the self-direction of students [4], increase their engagement [15] and show them how to apply programming concepts [16]. Like physical computing, live programming is based on the constructivist framework [12] supporting discovery [7] and active learning [20].

While mobile robots are often used by young, pre-literate children to learn simple aspects of coding such as sequencing, they typically don’t support more advanced concepts in computational thinking like repetition or variables. Our research seeks to fill this gap by exploring a robot-based physical computing solution that is live, portable, tangible, suitable for young children, and ultimately extendable and accessible to a diverse set of communities.

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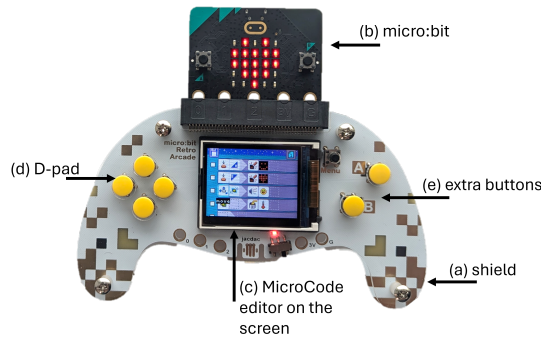


Figure 1: A micro:bit running MicroCode, plugged into a shield providing a colour display and extra button inputs.

2 MICROCODE - MICRO:BITS AND ROBOTS

MicroCode is a new portable programming system that enables coding of physical computing devices like the BBC micro:bit [9]. The MicroCode software and hardware provide a live, portable and visual programming environment. The micro:bit can be plugged into a portable battery powered shield, see Figure 1. These shields (a) have a slot for the micro:bit (b), a screen to view the MicroCode editor (c), a 4-way direction D-pad (d), extra buttons (e) and batteries. The shields can be used to program the micro:bit using MicroCode.

The MicroCode editor screen is based on a set of programming tiles, selected using the D-pad. These tiles, which children have referred to as "emojis" are fun and easy to recognise for pre-literate children. There is no text in MicroCode until a tile is selected, making it easier for children to understand. As shown in Figure 2, programming tiles are laid out in a manner many children are used to when learning to read: left-to-right and top-to-bottom. MicroCode allows more complex programs to be created than other devices for young children as it includes tiles for conditional execution, repetition, and variables. The tiles follow a **When:Do** pattern, with the inputs entered first under the **When** section on the left, and the output under the **Do** section on the right. The first rule in Figure 2 reads "**When** button A is pressed, **Do** draw a smiley face". The second and third row of Figure 2 shows some basic MicroCode commands for directing a remote robot car to move forward or stop based on user input. MicroCode uses live programming. Children can see their program on the screen and experience it running on the micro:bit in real time; as soon as a rule is written or amended, this is immediately reflected in the real world.

MicroCode has the ability to code a variety of robots by adding a second micro:bit slotted into a robot chassis, see Figure 3 (the robot is in the lower right of the picture). A standardised library was created that enables an ecosystem of robots to be controlled in this way. The micro:bit in the shield sends commands over the 2.4GHz radio to the micro:bit in the robot. In addition, MicroCode includes line following and distance sensing tiles. The sensors on the micro:bit robot send data back to the (MicroCode) micro:bit slotted in the shield.

In our preliminary studies, we found that using MicroCode with mobile robots resonates well with children and educators. The



Figure 2: A MicroCode program consisting of three When-Do rules. The first rule shows a happy face when the micro:bit's A button is pressed; the other two rules send commands to a wirelessly-connected mobile robot.



Figure 3: Controlling robots with MicroCode: two shields running MicroCode (upper-left) and a robot accessory using a separate micro:bit (lower-right).

freedom to create and amend code away from a host computer and see the effect of changes immediately works well in a robotics classroom where there is a lot of movement and need for constant changes. The portability of the MicroCode shield enables physical engagement and the small size of the hardware also encouraged collaboration and sharing of ideas. Educators noted how MicroCode altered students' perception. They were no longer just "learning to code", they were solving a real world problem using their robots.

3 CONCLUSION AND FUTURE WORK

MicroCode allows children of all ages and educators to program physical computing devices like the micro:bit and micro:bit robots in a live and portable way. MicroCode's live and easy to read programming model, as well as its portable shield that needs no mains electricity, a host computer or Internet, is a unique and novel way to introduce physical computing to new and younger audiences.

In the future we plan to formally evaluate MicroCode with the communities it aims to support. In particular we will run a study in which pre-literate children will use MicroCode-based robotics to understand the opportunities and limitations of our approach.

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